HONGYI YANG

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EDUCATION

Zhejiang University *Sep 2023 - Jun 2027 (expected)*

B.Eng. in Optoelectronic Information Science and Engineering; Chu Kochen Honors College (Mixed Class)

- **GPA:** 4.11 / 4.30 **Rank:** 1/103
- **TOEFL:** 103 (Reading 27 / Listening 27 / Speaking 25 / Writing 24)
- **Relevant Coursework:** Silicon Photonics: Technology and Devices (A+), Fundamentals and Applications of Quantum Optics (A+), Circuits and Analog Electronics (A+), Applied Optics (A), Electromagnetic Fields and Waves (A).

PUBLICATIONS & MANUSCRIPTS

- **Hongyi Yang**†, Haipeng Zhu†, Shiyu Feng, Zexi Lu, Xiyao Peng, Hongsheng Chen*, Sihan Zhao*, Dexin Ye*, and Haoliang Qian*. "Sub-Poissonian Electroluminescence from Resonant Inelastic Tunneling in One-Dimensional Gold Atomic Chains." *ACS Photonics*, 2026 (Published). *[DOI]*
- **Hongyi Yang**, Wendi Xia, Jun Li, Dexin Ye*, and Haoliang Qian*. "Physically Grounded Neural Networks: From Physical Substrates to Physics-Integrated Learning." *Applied Physics Reviews*, 2026 (Under review).

† equal contribution | * corresponding author

RESEARCH EXPERIENCE

Xu Research Group, Stanford University *Stanford, USA*

Visiting Undergraduate Research Intern; Advisor: Prof. Sheng Xu *Jul - Oct 2026 (on-site)*

- **Cleanroom Fabrication and Experimental Characterization**
  - Assisted with SiO_2 and TiO_2 thin-film deposition, photolithography, and lift-off processing.
  - Participated in ptychographic imaging of $FAPbI_3$ samples for structural characterization.
- **First-Principles Modeling of Ferroelectricity in $FAPbI_3$**
  - Built DFT models of epitaxially strained $FAPbI_3$ in Quantum ESPRESSO and performed phonon calculations to characterize the strained structures.
  - Mapped Pb off-centering and the associated double-well energy landscape, finding that compressive strain deepens the wells; linked Pb off-centering to polarization and identified the two minima as the origin of switchable ferroelectric states.

State Key Laboratory of Extreme Photonics and Instrumentation, Zhejiang University *Hangzhou, China*

Student Researcher; Advisor: Prof. Haoliang Qian *Apr 2025 - present*

- **Electrically Driven Non-Classical Light from Gold Atomic Chains**
  - Built a quantum-transport model of a gold atomic-chain junction from relativistic DFT, Wannier functions and DFPT vibrational modes, with bulk lead self-energies.
  - Solved the non-equilibrium Green's function problem without the wide-band approximation and treated electron-phonon coupling in the self-consistent Born approximation; implemented the pipeline in Python on HPC clusters.
  - Calculated current and current noise to identify sub-Poissonian electron statistics and studied how scattering and substitutional doping alter photon statistics.
  - Coupled the transport model to full-wave simulations of a plasmonic nanocube dimer to connect resonant inelastic tunneling with emitted light.

International Research Center for Advanced Photonics, Zhejiang University *Hangzhou, China*

Research Assistant; Advisor: Prof. Dawei Di *Sep 2023 - Jun 2024*

- **Perovskite Optoelectronic Devices**
  - Learned cleanroom processes for perovskite devices, including spin-coating and encapsulation.
  - Designed and built the optical setup used to measure external quantum efficiency.

SCHOLARSHIPS

2025-2026 Huawei Elite Scholarship
2024-2025 OPPO 'Benfen' Scholarship (8 recipients nationwide per year)
2023-2024 National Scholarship (top 0.2% nationwide)
2023-2024 First-Class Scholarship of Zhejiang University; Excellent Student Title

HONORS & AWARDS

2025-2026 First Prize, National Undergraduate Optoelectronic Design Competition
Oct 2025 Top Ten Undergraduates, School of Optoelectronic Science and Engineering, Zhejiang University
2024 First Prize, Zhejiang Provincial College Physics Competition
2023-2025 3rd Place (Long Jump), 5th Place (Triple Jump), Zhejiang University Sports Meeting
2022 First Prize, National High School Physics Olympiad (Hunan Province)

TECHNICAL SKILLS

Programming: Python, PyTorch, C/C++, LaTeX; high-performance computing (SLURM clusters)
Simulation: custom NEGF transport codes, Quantum ESPRESSO (DFT/DFPT), Wannier90, COMSOL Multiphysics, Zemax
Hardware: SolidWorks, Altium Designer

RESEARCH INTERESTS

Quantum photonics: integrated quantum photonics and on-chip quantum light sources
Device modelling: first-principles modelling of optoelectronic devices (NEGF, DFT/DFPT, Wannier functions)
Integration: hybrid integration of low-dimensional materials on silicon photonic platforms